Machine Learning-Based Credit Risk Prediction with Safe Loan Amount and EMI Recommendation

Financial institutions face increasing challenges in accurately evaluating the creditworthiness of loan applicants due to the growing complexity of financial data, class imbalance, and changing economic conditions. Traditional loan approval systems often depend on manual verification or basic statistical methods, making the process slow, inconsistent, and vulnerable to human error. As a result, high-risk borrowers may receive loans while genuinely eligible applicants may be rejected, leading to financial losses and an increase in non-performing loans (NPLs). Although recent machine learning models have significantly improved the accuracy of credit risk prediction, most existing systems focus only on identifying whether a customer is likely to default and do not provide practical decision support for financial institutions. They also lack transparency in explaining prediction results and fail to recommend an appropriate loan amount or affordable EMI based on the applicant's financial capacity. To overcome these limitations, this project proposes an Explainable Machine Learning-Based Credit Risk Prediction System that analyzes customer information such as income, employment status, credit history, existing debts, loan amount, repayment behavior, and other financial attributes to predict the probability of loan default. The system generates a risk score, explains the prediction using Explainable AI (SHAP), and recommends a safe loan amount and suitable EMI plan based on the applicant's repayment capability. This intelligent decision-support system aims to improve the accuracy, transparency, and efficiency of credit risk assessment, helping financial institutions make faster and more reliable lending decisions while reducing financial risk and providing customers with fair, explainable, and personalized loan recommendations.